BOT26105	BIOLOGY FOR COMPUTER ENGINEERS	CATEGORY	L	T	P	CREDIT
		BSC	3	0	0	3

Course Prerequisite / Preamble

- Basic understanding of Biology concepts studied at Higher Secondary level.
- This course provides the foundation for understanding biological systems and their engineering applications in biotechnology, healthcare, environment, and sustainable technologies.

Course Objective

- To introduce the fundamental concepts of biology relevant to engineering applications.
- To understand the structure and functions of cells and biomolecules.
- To familiarize students with genetics, biotechnology, and bioengineering concepts.
- To provide knowledge on biological processes and their industrial applications.
- To create awareness about environmental sustainability and biomedical innovations.

Course Outcome

CO1	Understand the structure and functions of cells and biomolecules.	**K2**
CO2	Explain biological processes and metabolic pathways relevant to engineering.	**K2**
CO3	Apply genetics and biotechnology concepts in engineering applications.	**K3**
CO4	Analyse environmental and biomedical applications of biology.	**K4**
CO5	Interpret the role of biological systems in sustainable engineering solutions	**K5**

Syllabus

Unit I	**Cell Biology**	**(9)**

Introduction to Biology – Cell as basic unit of life – Prokaryotic and eukaryotic cells – Cell structure and functions – Cell membrane – Nucleus – Mitochondria – Chloroplast – Cell division – Mitosis and meiosis – Applications in engineering and biotechnology.

Unit II	**Biomolecules and Enzymes**	**(9)**

Structure and functions of carbohydrates, proteins, lipids, and nucleic acids – DNA and RNA – Enzymes – Classification and mechanism of enzyme action – Factors affecting enzyme activity – Industrial and biomedical applications of enzymes.

Unit III	**Genetics and Biotechnology**	**(9)**
Principles of genetics – Mendel's laws – Chromosomes and genes – Genetic engineering – Recombinant DNA technology – Polymerase Chain Reaction (PCR) – Cloning – Applications of biotechnology in medicine, agriculture, and industry.		
Unit IV	**Microbiology and Environmental Biology**	**(9)**
Introduction to microorganisms – Bacteria, fungi, and viruses – Microbial growth – Industrial microbiology – Waste management – Bioremediation – Environmental pollution and control – Sustainable environmental practices.		
Unit V	**Bioengineering and Biomedical Applications**	**(9)**
Introduction to bioengineering – Tissue engineering – Biosensors – Biomedical devices – Artificial organs – Bioinformatics – Applications of biology in healthcare engineering – Ethical issues in biotechnology and biomedical engineering.		

Total Hours: 45

Text Books

1. T. Johnson, Biology for Engineers, CRC Press, 2nd Edition (2018).
2. Campbell Biology, by Campbell & Reece, now continued by Urry, Minorsky, Hull, and Orr, Pearson Education, 13th Edition.
3. U. Satyanarayana, Biotechnology, Books and Allied Publications, 3rd Edition (2020).

Reference

1. Lehninger, Principles of Biochemistry, W.H. Freeman Publications.
2. Prescott, Microbiology, McGraw Hill Education.
3. B.D. Singh, Biotechnology, Kalyani Publishers.
4. Gerard J. Tortora, Principles of Anatomy and Physiology, Wiley Publications.

Web / NPTEL Resources

1. https://nptel.ac.in/courses/121106008
2. https://nptel.ac.in/courses/102106081